

SECURITY INCIDENT ANALYSIS REPORT

PowerShell-Based Multi-Stage Attack Chain Investigation

Report ID	SOC-PS-2026-002
Report Date	April 23, 2026
Prepared By	Mehrab Shaheen
Classification	CONFIDENTIAL — TLP:RED — Restricted Distribution
SIEM Platform	Wazuh v4.x
Log Sources	Sysmon (Event ID 1, 11, 13), Windows Security (1102)
Endpoint	DESKTOP-KA2LJHB IP: 192.168.x.x
Attack Type	PowerShell-Based Multi-Stage Attack (Execution + Persistence + Evasion)
Threat Level	HIGH — True Positive — Complete Attack Chain Detected
Status	INVESTIGATION COMPLETE — Remediation Pending

1. EXECUTIVE OVERVIEW

- During analysis of endpoint DESKTOP-KA2LJHB (192.168.x.x), a sequence of security events was detected and investigated that revealed a deliberate, multi-stage attack simulation targeting the Windows endpoint. The investigation identified six correlated alerts spanning PowerShell-based payload deployment, execution policy bypass, SYSTEM-level privilege escalation, registry persistence, automated re-execution, and Security log tampering.
- The activity is not a collection of isolated anomalies. Each alert is a logical consequence of the previous one, forming a complete attacker workflow from initial staging through to evidence destruction. The attack demonstrates clear adversarial intent across three primary MITRE ATT&CK categories: Execution, Persistence, and Defense Evasion.

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Attack Stages

- **Stage 1: Payload Creation:** PowerShell drops backdoor.ps1 in C:\Users\Public (Sysmon Event ID 11)
- **Stage 2: Script Execution:** backdoor.ps1 executed with ExecutionPolicy Bypass parameter (Sysmon Event ID 1)
- **Stage 3: SYSTEM Artifact:** PSScriptPolicyTest file created in C:\Windows\Temp — confirms SYSTEM-level execution
- **Stage 4: Registry Persistence:** Run key modified via reg.exe — ensures script auto-executes at login (Sysmon Event ID 13)
- **Stage 5: Automated Execution:** svchost.exe spawns PowerShell — persistence mechanism confirmed active
- **Stage 6: Defense Evasion:** Windows Security logs cleared (Event ID 1102) — forensic evidence destroyed locally

2. Chronological Attack Flow

- *Each stage reinforces the previous one. No event is accidental or isolated — the temporal and logical dependency chain confirms a scripted, deliberate attack workflow:*

Stage	Event	Sysmon / Event ID	Chain Role
1	Payload Creation	Sysmon Event ID 11	Drops backdoor.ps1 in Public dir — enables all subsequent steps

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Stage	Event	Sysmon / Event ID	Chain Role
2	Script Execution	Sysmon Event ID 1	Executes payload with -ExecutionPolicBypass — bypasses security controls
3	SYSTEM Artifact	Sysmon Event ID 11	PSScriptPolicyTest confirms execution escalated to SYSTEM context
4	Registry Persistence	Sysmon Event ID 13	Run key set — payload survives reboots and auto-executes at login
5	Automated Execution	Process Chain (svchost)	Svchost.exe spawns PowerShell — confirms persistence is live and active
6	Log Clearing	Windows Event ID 1102	Security logs deleted — local forensic evidence destroyed

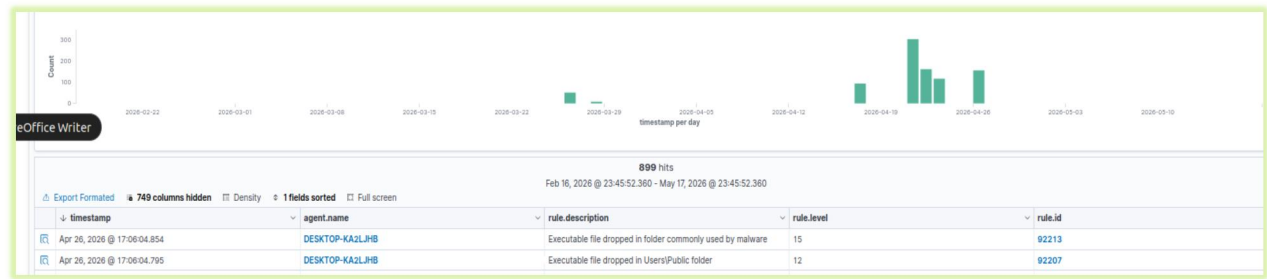
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3. DETAILED ALERT ANALYSIS

- Each alert is analyzed using the 5W methodology, supported by technical evidence extraction, behavioral deep-analysis, and individual verdict classification.

Alert 1 — Payload Creation | Sysmon Event ID 11 (File Create)



Document Details		View surrounding documents	View single document
Table	JSON		
f _index	wazuh-alerts-4.x-2026.04.26		
f agent.id	005		
f agent.ip			
f agent.name	DESKTOP-KA2LJHB		
f data.win.eventdata.creationUtcTime	2026-04-23 09:57:01.435		
f data.win.eventdata.image	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe		
f data.win.eventdata.processGuid	{faf25ee8-ec74-69e9-9802-000000003d00}		
f data.win.eventdata.processId	944		
f data.win.eventdata.targetFileName	C:\Users\Public\backdoor.ps1		
f data.win.eventdata.user	DESKTOP-KA2LJHB\mehrab		
f data.win.eventdata.utcTime	2026-04-23 09:57:01.435		
f data.win.system.channel	Microsoft-Windows-Sysmon/Operational		
f data.win.system.computer	DESKTOP-KA2LJHB.lab.local		
f data.win.system.eventID	11		
f data.win.system.eventRecordID	60480		
f data.win.system.keywords	0x8000000000000000		
f data.win.system.level	4		
f data.win.system.message	"File created: RuleName: - UtcTime: 2026-04-23 09:57:01.435 ProcessGuid: {faf25ee8-ec74-69e9-9802-000000003d00} ProcessId: 944 Image: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe TargetFileName: C:\Users\Public\backdoor.ps1 CreationUtcTime: 2026-04-23 09:57:01.435 User: DESKTOP-KA2LJHB\mehrab"		
f data.win.system.opcode	0		
f data.win.system.processID	2428		
f data.win.system.providerGuid	{5770385f-c22a-43e0-bf4c-06f5698ffbd9}		
f data.win.system.providerName	Microsoft-Windows-Sysmon		
f data.win.system.severityValue	INFORMATION		
f data.win.system.systemTime	2026-04-23T09:57:01.4462383Z		

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Element	Detail
Who	User mehrab — PowerShell process acting under user context
What	PowerShell created backdoor.ps1 — an executable script file
When	Initial stage — first recorded activity in the attack chain
Where	C:\Users\Public\ — shared, writable, commonly overlooked directory
Why	Pre-positioning payload for later execution — classic staging behavior

Technical Details

Id	Value
Sysmon Event ID	11 — FileCreate
File Created	C:\Users\Public\backdoor.ps1
Creating Process	powershell.exe
User Context	mehrab
File Extension	.ps1 — PowerShell executable script
Directory Risk	U sers\Public — writable by standard users, frequently unmonitored

Technical Analysis

- PowerShell used as a file creation tool — not its normal use case — indicates script-based automation rather than manual file placement
- The .ps1 extension marks the file as an executable scripting payload, not a configuration or log file

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- C:\Users\Public is a high-risk staging location: writable by non-admin users, commonly excluded from security monitoring policies, and frequently used in malware staging
- No legitimate software installation or administrative workflow creates .ps1 scripts in the Public directory without documented change management
- This file's creation is the enabling dependency for every subsequent alert — without it, no execution, persistence, or evasion occurs

Analyst Interpretation

- The creation of backdoor.ps1 in C:\Users\Public by PowerShell is not incidental. This is deliberate payload pre-positioning — the attacker has written an executable script to a location that is both accessible and under-monitored. The use of PowerShell as the file creation mechanism indicates automation or remote script delivery rather than manual attacker interaction. This event should be treated as the beginning of a structured attack chain, not a standalone anomaly.

VERDICT — TRUE POSITIVE

True Positive — Malicious Payload Staging. This is the foundational event that enables the entire subsequent attack chain.

Recommendations

- Enable Sysmon Event ID 11 monitoring with alerts on .ps1 file creation outside of approved administrator workflows
- Alert specifically on PowerShell.exe creating files in user-writable directories (Desktop, Public, Temp, AppData)

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- Implement Software Restriction Policy (SRP) or AppLocker to block .ps1 execution from C:\Users\Public
- Flag any new .ps1 file appearing outside C:\Program Files or system script directories for immediate review

Alert 2 — PowerShell Script Execution | Sysmon Event ID 1 (Process Create)

data.win.eventdata.sessionId	1
data.win.eventdata.user	DESKTOP-KA2LJHB\mehrab
data.win.eventdata.utcTime	2026-04-26 13:31:01.930
data.win.system.channel	Microsoft-Windows-Sysmon/Operational
data.win.system.computer	DESKTOP-KA2LJHB.lab.local
data.win.system.eventID	1
data.win.system.eventRecordID	61270
data.win.system.keywords	0x8000000000000000
data.win.system.level	4
data.win.system.message	"Process Create: RuleName: - UtcTime: 2026-04-26 13:31:01.930 ProcessGuid: {faf25ee8-15-69ee-1e05-000000003d00} ProcessId: 4380 Image: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe FileVersion: 10.0.19041.3636 (WinBuild.160101.0800) Description: Windows PowerShell Product: Microsoft® Windows® Operating System Company: Microsoft Corporation OriginalFileName: PowerShell.EXE CommandLine: "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.EXE" -ExecutionPolicy Bypass -File C:\Users\Public\backdoor.ps1 ParentDirectory: C:\Windows\System32\ User: DESKTOP-KA2LJHB\mehrab ProcessGuid: {faf25ee8-15-69ee-1e05-000000003d00}
data.win.system.opcode	0
data.win.system.processID	2428
data.win.system.providerGuid	{5770385f-c22a-43e0-bf4c-06f5698ffbd9}
data.win.system.providerName	Microsoft-Windows-Sysmon
data.win.system.severityValue	INFORMATION
data.win.system.systemTime	2026-04-26T13:31:01.9576518Z
data.win.system.task	1
data.win.system.threadID	3548
data.win.system.version	5
decoder.name	windows_eventchannel
id	1777227227.3883062
input.type	log

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5W Analysis

Element	Detail
Who	User mehrab — executing previously staged payload
What	PowerShell launched backdoor.ps1 with ExecutionPolicy Bypass
When	Immediately after payload creation in Alert 1
Where	Local system — same endpoint
Why	Active exploitation — executing staged payload while bypassing security controls

Technical Details

Field	Value
Sysmon Event ID	1 — ProcessCreate
Process Launched	powershell.exe
User Context	mehrab
Command Observed	powershell -ExecutionPolicy Bypass -File C:\Users\Public\backdoor.ps1
-ExecutionPolicy	By pass — explicitly disables PowerShell execution security policy
-File	Direct execution of external pre-written script

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Observed Command

```
powershell.exe -ExecutionPolicy Bypass -File C:\Users\Public\backdoor.ps1
```

Technical Analysis

- -ExecutionPolicy Bypass is a deliberate security override, it forces PowerShell to run scripts that would otherwise be blocked by organizational execution policy
- Legitimate system administrators use constrained or AllSigned execution policies; Bypass is exclusively used to circumvent those controls
- The -File parameter specifies an externally pre-written script — confirming this is planned, not interactive attacker input
- The script path (C:\Users\Public\backdoor.ps1) directly links this execution event to the file creation in Alert 1 — confirming single actor, single session
- This combination of parameters (Bypass + external script file) has extremely limited legitimate use and is widely recognized as a malicious execution pattern

Analyst Interpretation

- The execution of backdoor.ps1 with **-ExecutionPolicy Bypass** is unambiguous evidence of malicious intent. The attacker is not working within the security boundaries of the system — they are explicitly dismantling them. The direct link between this event and Alert 1 (same file path, same user, consecutive timing) confirms this is stage two of a pre-planned attack sequence. This event transitions the attack from staging to active execution.

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VERDICT — TRUE POSITIVE

T True Positive — Malicious Execution. ExecutionPolicy Bypass with external script is a confirmed high-confidence attack indicator.

Recommendations

- Enable PowerShell Script Block Logging (Event ID 4104) to capture full script content at execution time
- Create Wazuh rule: alert HIGH when powershell.exe is launched with -ExecutionPolicy Bypass parameter
- Restrict script execution via Group Policy: set PowerShell Execution Policy to AllSigned or RemoteSigned for all endpoints
- Implement Constrained Language Mode in PowerShell for non-administrator users

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Alert 3 — SYSTEM-Level Execution Artifact | Sysmon Event ID 11

Document Details		View surrounding documents	View single document
Table	JSON		
index	wazuh-alerts-4.x-2026.04.26		
agent.id	005		
agent.ip			
agent.name	DESKTOP-KA2LJHB		
data.win.eventdata.creationUtcTime	2026-04-26 13:22:28.037		
data.win.eventdata.image	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.EXE		
data.win.eventdata.processGuid	{faf25ee8-1179-69ee-0f05-000000003d00}		
data.win.eventdata.processId	4428		
data.win.eventdata.targetFilename	C:\Users\mehrab\AppData\Local\Temp_PSScriptPolicyTest_et3m05tt.4vx.ps1		
data.win.eventdata.user	DESKTOP-KA2LJHB\mehrab		
data.win.eventdata.utcTime	2026-04-26 13:22:28.037		
data.win.system.channel	Microsoft-Windows-Sysmon/Operational		
data.win.system.computer	DESKTOP-KA2LJHB\lab.local		
data.win.system.eventId	11		
data.win.system.eventRecordId	61265		
data.win.system.keywords	0x8000000000000000		
data.win.system.level	4		
data.win.system.message	"File created: RuleName: - UtcTime: 2026-04-26 13:22:28.037 ProcessGuid: {faf25ee8-1179-69ee-0f05-000000003d00} ProcessId: 4428 Image: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.EXE TargetFilename: C:\Users\mehrab\AppData\Local\Temp_PSScriptPolicyTest_et3m05tt.4vx.ps1 CreationUtcTime: 2026-04-26 13:22:28.037 User: DESKTOP-KA2LJHB\mehrab"		
data.win.system.opcode	0		
data.win.system.processId	2428		
data.win.system.providerGuid	{5770385f-c22a-43e0-bf4c-06f5698ffb9}		
data.win.system.providerName	Microsoft-Windows-Sysmon		
data.win.system.severityValue	INFORMATION		
data.win.system.systemTime	2026-04-26T13:22:28.0538952Z		
data.win.system.task	11		

5W Analysis

Element	Detail
Who	NT AUTHORITY\SYSTEM — highest privilege account on Windows
What	PowerShell created a temporary policy test script (__PSScriptPolicyTest)
When	During or immediately after script execution in Alert 2
Where	C:\Windows\Temp — system-level temporary directory

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Element	Detail
Why	Reveals script execution occurred under SYSTEM context — critical privilege indicator

Technical Details

Field	Value
Sysmon Event ID	11 — FileCreate
File Created	C:\Windows\Temp__PSScriptPolicyTest_*.ps1
Creating Process	powershell.exe
User Context	NT AUTHORITY\SYSTEM
File Origin	Auto-generated by PowerShell during script policy validation
Significance	Presence confirms execution occurred at SYSTEM privilege level

Technical Analysis

- The __PSScriptPolicyTest file is an internal PowerShell artifact — automatically generated during execution policy validation, not attacker-placed
- Its significance is entirely contextual: its presence in C:\Windows\Temp under NT AUTHORITY\SYSTEM proves that PowerShell executed a script at SYSTEM privilege level

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- Standard users do not trigger SYSTEM-level PowerShell execution through normal activity — this points to a scheduled task, service, or privileged automation mechanism
- This artifact bridges Alert 2 (user-initiated execution) and Alert 5 (automated SYSTEM execution) — it is evidence that the execution chain had already transitioned to SYSTEM context
- SYSTEM-level execution means the payload had unrestricted access to the entire system: all files, registry hives, services, and security functions

Analyst Interpretation

This alert is the most technically subtle in the chain, but one of the most significant. The `_PSScriptPolicyTest` artifact is not attacker-created, it is a Windows-generated file that inadvertently exposes the execution context. Its presence in `C:\Windows\Temp` under SYSTEM context is definitive proof that the `backdoor.ps1` script was executed with full SYSTEM privileges, not merely under the user 'mehrab'. This confirms that the attacker achieved privilege escalation — whether through a scheduled task, a service, or another elevation mechanism.

VERDICT — CRITICAL

True Positive — Execution Context Escalation Confirmed. SYSTEM-level PowerShell execution is a critical high-severity indicator.

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Recommendations

- Monitor all SYSTEM-level PowerShell process creation events — alert immediately when powershell.exe runs under NT AUTHORITY\SYSTEM
- Correlate __PSScriptPolicyTest file creation events with parent process trees to identify the execution trigger
- Audit all scheduled tasks and services for PowerShell-based commands — especially those referencing C:\Users\Public
- Implement SYSTEM-level process execution alerting in Wazuh as a Tier 1 priority alert.

Alert 4 — Registry Persistence | Sysmon Event ID 13 (Registry Value Set)

Document Details		View surrounding documents	View single document
Table	JSON		
index	wazuh-alerts-4.x-2026.04.26		
agent.id	005		
agent.ip			
agent.name	DESKTOP-KA2LJHB		
data.win.eventdata.details	powershell -ExecutionPolicy Bypass -File C:\Users\Public\backdoor.ps1		
data.win.eventdata.eventType	SetValue		
data.win.eventdata.image	C:\Windows\system32\reg.exe		
data.win.eventdata.processGuid	{faf25ee8-0a9d-69ee-e704-000000003d00}		
data.win.eventdata.processId	8436		
data.win.eventdata.ruleName	T1060,RunKey		
data.win.eventdata.targetObject	HKU\{S-1-5-21-1410036601-668628055-412612306-1001}\SOFTWARE\Microsoft\Windows\CurrentVersion\Run\updater		
data.win.eventdata.user	DESKTOP-KA2LJHB\mehrab		
data.win.eventdata.utcTime	2026-04-26 12:52:45.428		
data.win.system.channel	Microsoft-Windows-Sysmon/Operational		
data.win.system.computer	DESKTOP-KA2LJHB.lab.local		
data.win.system.eventId	13		
data.win.system.eventRecordId	61206		
data.win.system.keywords	0x8000000000000000		
data.win.system.level	4		
data.win.system.message	"Registry value set: RuleName: T1060,RunKey EventType: SetValue UtcTime: 2026-04-26 12:45.428 ProcessGuid: {faf25ee8-0a9d-69ee-e704-000000003d00} ProcessId: 8436 Image: \Windows\system32\reg.exe TargetObject: HKU\{S-1-5-21-1410036601-668628055-412612: 6-1001}\SOFTWARE\Microsoft\Windows\CurrentVersion\Run\updater Details: powershell -Ex tionPolicy Bypass -File C:\Users\Public\backdoor.ps1 User: DESKTOP-KA2LJHB\mehrab"		

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5W Analysis

Element	Detail
Who	User mehrab — using reg.exe to modify the Windows Registry
What	Run key modified to auto-execute backdoor.ps1 at every user login
When	After execution in Alerts 2 and 3 — persistence establishment phase
Where	HKCU\Software\Microsoft\Windows\CurrentVersion\Run\updater
Why	Ensures the payload survives reboots and re-executes automatically — persistent access

Technical Details

Field	Value
Sysmon Event ID	13 — RegistryEvent (Value Set)
Registry Path	HKCU\Software\Microsoft\Windows\CurrentVersion\Run\updater
Registry Value	powershell -ExecutionPolicy Bypass -File C:\Users\Public\backdoor.ps1
Process Used	reg.exe — Windows Registry editor command-line tool
Key Name	'updater' — generic name chosen to blend with legitimate software
Persistence Type	User-level Run key — executes at login for account 'mehrab'

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Registry Entry Set

Key: HKCU\Software\Microsoft\Windows\CurrentVersion\Run

Name: updater

Value: powershell -ExecutionPolicy Bypass -File C:\Users\Public\backdoor.ps1

Technical Analysis

- HKCU\Run is a classic Windows persistence location — every value in this key is automatically executed when the user logs in
- The key name 'updater' is deliberately generic — designed to appear as a legitimate software update mechanism during cursory registry inspection
- Instead of a binary executable, the attacker uses a PowerShell command — this avoids dropping a new executable on disk, which would be more likely to trigger AV detection
- The embedded command reuses the exact same -ExecutionPolicy Bypass -File backdoor.ps1 pattern from Alert 2 — confirming a single actor maintaining the same toolchain
- This persistence entry ensures the backdoor survives system reboots, user re-logins, and basic incident response that does not include registry auditing

Analyst Interpretation

- The registry Run key modification is the centerpiece of this attack's persistence strategy. The attacker has ensured that even if the investigation results in a simple password reset or account lockout, the payload will continue to execute each time the user 'mehrab' logs in. The deliberate use of a trusted binary (PowerShell), a trusted registry location (Run key), and an

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innocuous-sounding key name ('updater') reflects sophisticated attacker tradecraft aimed at surviving both technical detection and administrative response.

VERDICT — CRITICAL

CRITICAL — Registry Persistence Confirmed. Automated re-execution guaranteed at every user login until manually remediated.

Recommendations

- Immediately delete the registry value:
HKCU\Software\Microsoft\Windows\CurrentVersion\Run\updater
- Monitor all Run key modifications via Sysmon Event ID 13 — alert on any PowerShell command embedded in Run key values
- Create Wazuh rule: CRITICAL alert when reg.exe writes to any CurrentVersion\Run or CurrentVersion\RunOnce path
- Perform a full registry audit for persistence keys: Run, RunOnce, RunOnceEx, Winlogon, Services, Scheduled Tasks
- Implement periodic automated registry baseline comparison to detect unauthorized persistence additions

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Alert 5 — Automated SYSTEM Execution | svchost.exe → powershell.exe

Document Details		View surrounding documents	View single document
Table	JSON		
index	wazuh-alerts-4.x-2026.04.26		
agent.id	005		
agent.ip			
agent.name	DESKTOP-KA2LJHB		
data.win.eventdata.commandLine	"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.EXE" -ExecutionPolicy Bypass -File C:\Users\Public\backdoor.ps1		
data.win.eventdata.company	Microsoft Corporation		
data.win.eventdata.currentDirectory	C:\Windows\system32\		
data.win.eventdata.description	Windows PowerShell		
data.win.eventdata.fileVersion	10.0.19041.3636 (WinBuild.160101.0800)		
data.win.eventdata.hashes	MD5=6726185B70B5ADF05E8A1A1DF82EBF30,SHA256=64DD55E1C2373DEED25C2776F553C632E58C45E56A0E4639DFD54EE97EAB9C19,IMPHASH=E3007C8E0098D06ABF617EEE6F0C5ABD		
data.win.eventdata.image	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe		
data.win.eventdata.integrityLevel	Medium		
data.win.eventdata.logonGuid	{faf25ee8-cddf-69e9-63cb-080000000000}		
data.win.eventdata.logonId	0x8cb63		
data.win.eventdata.originalFileName	PowerShell.EXE		
data.win.eventdata.parentCommandLine	C:\Windows\system32\svchost.exe -k netsvcs -p		
data.win.eventdata.parentImage	C:\Windows\System32\svchost.exe		
data.win.eventdata.parentProcessGuid	{faf25ee8-cdc6-69e9-1400-000000003d00}		
data.win.eventdata.parentProcessId	448		
data.win.eventdata.parentUser	NT AUTHORITY\SYSTEM		

5W Analysis

Element	Detail
Who	NT AUTHORITY\SYSTEM — automated, non-interactive execution
What	svchost.exe spawned a PowerShell process — background automated

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Element	Detail
	execution
When	After registry persistence (Alert 4) — persistence mechanism triggered
Where	Local system — background service execution context
Why	Confirms persistence is live — payload now executes automatically without user interaction

Technical Details

Field	Value
Process Chain	svchost.exe → powershell.exe
Parent Process	svchost.exe (Windows Service Host — system process)
Child Process	powershell.exe (spawned by svchost — highly abnormal)
Execution Context	NT AUTHORITY\SYSTEM — non-interactive background execution
User Interaction	None — fully automated execution
Persistence Linkage	Directly validates the registry Run key established in Alert 4

Technical Analysis

- svchost.exe is a Windows system process that hosts Windows services — it does not legitimately spawn user-interactive applications like PowerShell

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- When svchost.exe is observed as the parent of powershell.exe, this is a strong indicator of background, service-triggered, or scheduled execution — a hallmark of persistence mechanisms
- The SYSTEM execution context confirms this is not a user-initiated action — the payload is executing autonomously, as intended by the registry Run key setup
- This parent-child process relationship (svchost → PowerShell) is a well-documented indicator of attacker persistence being actively exercised
- The automated nature of this execution means the backdoor will continue running in the background with SYSTEM privileges until the persistence mechanism is removed

Analyst Interpretation

- Alert 5 is the operational proof that the persistence established in Alert 4 is working. The registry Run key has triggered PowerShell via the Windows service infrastructure, executing the backdoor under SYSTEM privileges without any user interaction. This alert confirms the attack has reached a self-sustaining state — the payload will continue to execute automatically at defined intervals or login events. This is the most operationally dangerous phase of the chain, as the attacker no longer needs to be actively present to maintain access.

VERDICT — CRITICAL

CRITICAL — Active Automated Backdoor Confirmed. Persistence mechanism is live.
Immediate endpoint isolation required.

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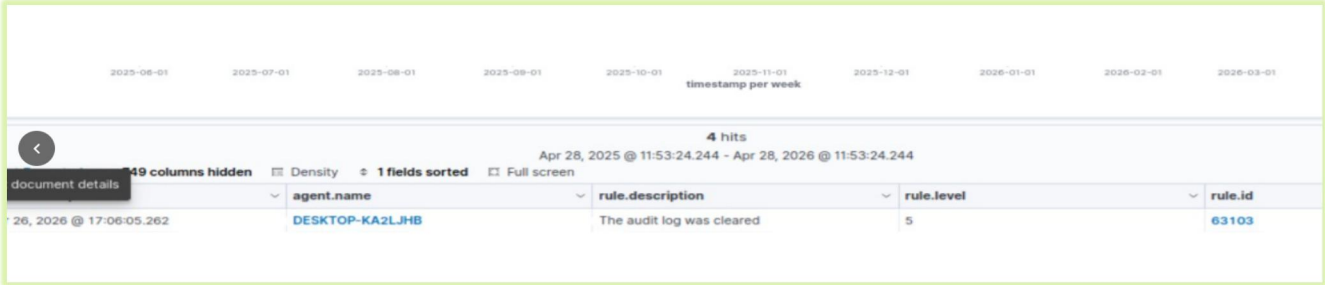
Recommendations

- Immediately isolate DESKTOP-KA2LJHB from the network to prevent further automated payload execution
- Alert on svchost.exe spawning PowerShell, cmd.exe, wscript.exe, or cscript.exe — this is always abnormal behavior
- Investigate all services running under SYSTEM context that could be triggering the PowerShell execution
- Terminate the running powershell.exe process and remove the registry persistence key before restarting the endpoint.

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Alert 6 — Security Log Cleared | Windows Event ID 1102 — Defense Evasion



agent.name	DESKTOP-KA2LJHB
data.win.logFileCleared.clientProcessId	2856
data.win.logFileCleared.clientProcessStartKey	17169973579350700
data.win.logFileCleared.subjectDomainName	DESKTOP-KA2LJHB
data.win.logFileCleared.subjectLogonId	0x8cab0
data.win.logFileCleared.subjectUserName	mehrab
data.win.logFileCleared.subjectUserSid	S-1-5-21-1410036601-668628055-412612306-1001
data.win.system.channel	Security
data.win.system.computer	DESKTOP-KA2LJHB.iab.local
data.win.system.eventID	1102
data.win.system.eventRecordID	94132
data.win.system.keywords	0x4020000000000000
data.win.system.level	4
data.win.system.message	"The audit log was cleared. Subject: Security ID: S-1-5-21-1410036601-668628055-412612306-1001 Account Name: mehrab Domain Name: DESKTOP-KA2LJHB Logon ID: 0x8CAB0"
data.win.system.opcode	0
data.win.system.processID	964
data.win.system.providerGuid	{fc65ddd8-d6ef-4962-83d5-6e5cfe9ce148}
data.win.system.providerName	Microsoft-Windows-Eventlog
data.win.system.severityValue	INFORMATION
data.win.system.systemTime	2026-04-23T10:09:37.0135470Z
data.win.system.task	104
data.win.system.threadID	5076

SECURITY INCIDENT ANALYSIS REPORT

PowerShell-Based Multi-Stage Attack Chain Investigation

5W Analysis

Element	Detail
Who	User mehrab — Administrator privilege confirmed by the action itself
What	Windows Security Event Log completely cleared — all authentication and audit events deleted
When	Final stage — after all attack objectives achieved
Where	Windows Security Event Log on DESKTOP-KA2LJHB
Why	Destroying local forensic evidence — preventing investigation of the complete attack chain

Technical Details

Field	Value
Windows Event ID	1102 — The audit log was cleared
Log Type Cleared	Windows Security Event Log
Privilege Required	SecurityPrivilege or local Administrator membership
Acting User	mehrab
Events Erased	All Sysmon events, auth logs, registry events, process creation — everything local
SIEM Retention	Wazuh had already forwarded all events — local clearing has zero impact on SIEM records

SECURITY INCIDENT ANALYSIS REPORT

PowerShell-Based Multi-Stage Attack Chain Investigation

Technical Analysis

- Clearing the Windows Security log requires SeSecurityPrivilege — this action retroactively confirms that mehrab (or the process acting on their behalf) held Administrator-level privileges throughout the attack
- Event ID 1102 is generated at the exact moment of log clearing — in an endpoint-only scenario, this would be the single remaining trace of the entire attack
- All prior events (Sysmon 11, 1, 13; process chains; authentication events) are destroyed from the local machine — a complete forensic wipe
- This action follows the completion of all attack objectives — execution done, persistence established, automation confirmed — indicating the attacker is performing planned cleanup
- Because Wazuh SIEM forwards events in real time, the log clearing has zero forensic impact on the centrally retained evidence — every alert in this chain is preserved

Analyst Interpretation

The clearing of the Windows Security log is the final act of an operationally aware threat actor. By erasing local logs, the attacker intended to prevent any forensic investigation that relies on endpoint-stored evidence. This behavior demonstrates that the attacker understood they were being — or could be — monitored. The fact that Event ID 1102 itself survived (because Wazuh had already ingested it) exemplifies why centralized SIEM log retention is a non-negotiable security control. Without Wazuh, this investigation would have had no evidentiary basis.

VERDICT — CRITICAL

CRITICAL — Defense Evasion Confirmed. Log clearing is always malicious in context. Retroactively confirms full administrator-level compromise.

SECURITY INCIDENT ANALYSIS REPORT

PowerShell-Based Multi-Stage Attack Chain Investigation

Recommendations

- Configure Wazuh Active Response to automatically isolate the endpoint upon detection of Event ID 1102
- Restrict the Security log clearing privilege (SeSecurityPrivilege) via GPO — remove from all accounts except the dedicated audit service account
- Alert on Event ID 1102 as an automatic P1/Critical incident — no investigation needed, immediate response required
- Ensure all Windows endpoints forward Security, System, and Sysmon logs to Wazuh in real time — local log clearing must have zero forensic impact.